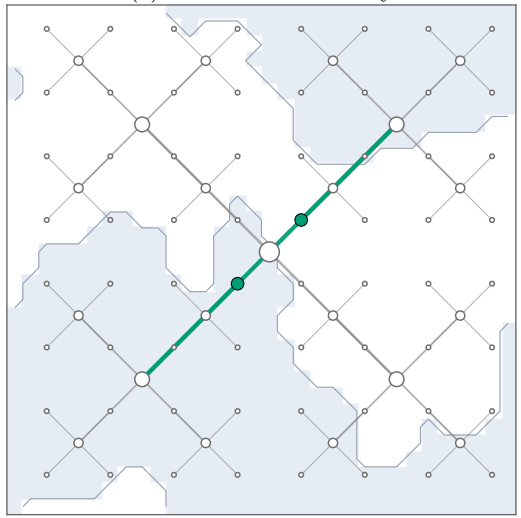
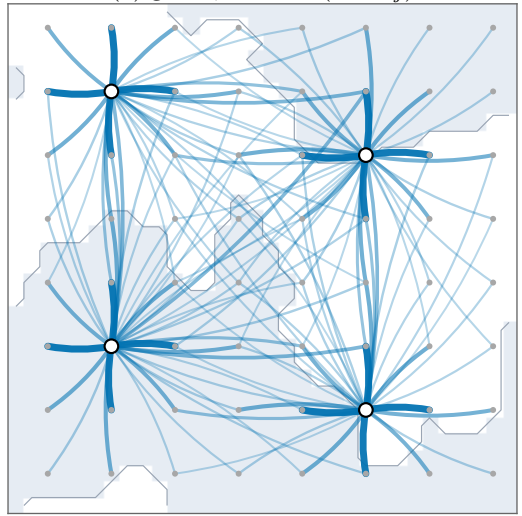
(a) tree: U-Net hierarchy



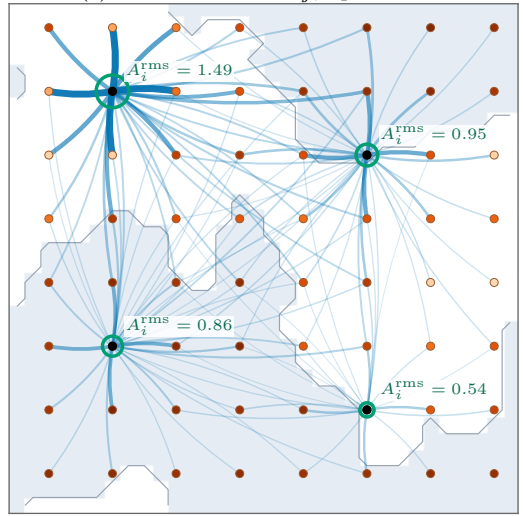
neighbouring cells across a block
boundary meet only at the root

(b) global, fixed: $\kappa_\theta(\mathbf{r}_i - \mathbf{r}_j)$



every cell hears every other through
the same pattern (Fourier layer)

(c) rhizome: $\mathbf{A}_i\,\kappa_\theta\,\mathbf{B}_j$, update t=4



edge $j \to i$ scaled by the receiver gate
$\mathbf{A}_i$ (ring) and the sender gate $\mathbf{B}_j$ (shade)